

Q Q̄ Decay

$\Phi = s\bar{s}$ resonance

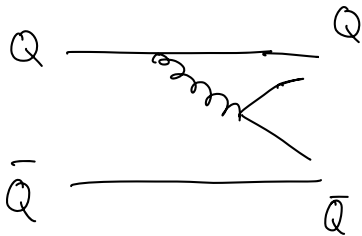
$$M \approx 1020 \text{ MeV}$$

$$\Gamma = 4 \text{ MeV}$$

$\rho = u\bar{u}, d\bar{d}$ resonance

$$M = 770 \text{ MeV}$$

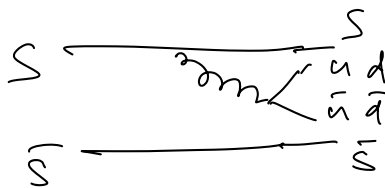
$$\Gamma = 150 \text{ MeV}$$



$$\rho \rightarrow \pi^+\pi^-, \pi^0\pi^0$$

$$Q = m_\rho - 2m_\pi = 770 - 2 \times 140 = 600 \text{ MeV}$$

$\Rightarrow \rho$ wide resonance



$$\Phi \rightarrow K^+K^-, K^0\bar{K}^0$$

$$Q = 1020 - 2m_K \approx 20 \text{ MeV}$$

$$m_K \approx 498 \text{ MeV}$$

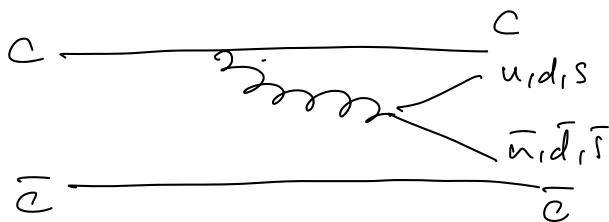
Dominant decay $\Phi \rightarrow K\bar{K}$
has a small phase space

$\phi(1020)$ DECAY MODES

Mode	Fraction (Γ_i/Γ)	Scale factor/ Confidence level
$\Gamma_1 K^+K^-$	(49.1 $\pm$ 0.5) %	S=1.3
$\Gamma_2 K_L^0\bar{K}_S^0$	(33.9 $\pm$ 0.4) %	S=1.2
$\Gamma_3 \rho\pi + \pi^+\pi^-\pi^0$	(15.4 $\pm$ 0.4) %	S=1.2
$\Gamma_4 \rho\pi$		
$\Gamma_5 \pi^+\pi^-\pi^0$		
$\Gamma_6 \eta\gamma$	(1.301 $\pm$ 0.025) %	S=1.2
$\Gamma_7 \pi^0\gamma$	(1.32 $\pm$ 0.05) $\times 10^{-3}$	
$\Gamma_8 \ell^+\ell^-$	—	

$J/\psi = c\bar{c}$ resonance

$$M = 3100 \text{ MeV} \quad \Gamma = 0.087 \text{ MeV}$$



$$J/\psi \rightarrow (c\bar{u}) + (c\bar{d}) + (c\bar{s}) + (\bar{c}u) + (\bar{c}d) + (\bar{c}s)$$

$$c\bar{u} = D^0 \quad m_{D^0} \approx 1.864 \text{ GeV}$$

$$\bar{c}u = \bar{D}^0$$

$$J/\psi \rightarrow D^0\bar{D}^0$$

$$Q = 3.1 - 2 \times 1.864 < 0$$

Kinematically not allowed.

$$c\bar{d} = D^+$$

$$Q < 0$$

$$\bar{c}d = D^-$$

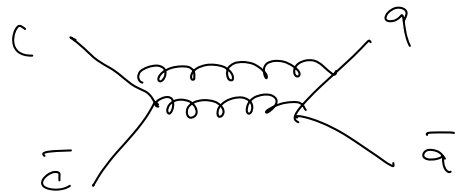
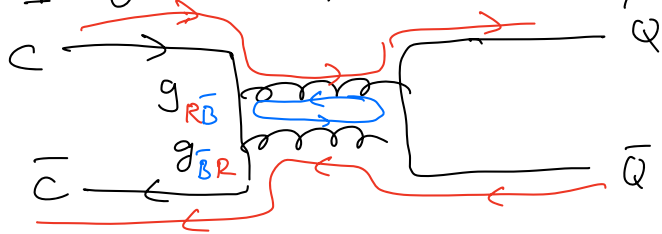
$$c\bar{s} = D_s^+$$

$$\bar{c}s = D_s^-$$

$$D_s^+$$

$$Q < 0$$

1 gluon not allowed by every conservation.



$$Q: (c\bar{c}) \quad J^{PC} = 1^{--}$$

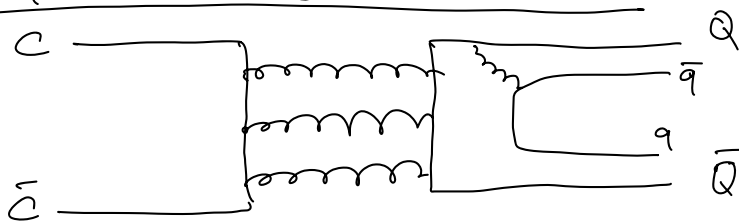
$$Q = (-1)^{L+I+S} = (-1)^{0+0+1} = -1 \quad \text{Same as } \gamma$$

$$J/\psi \rightarrow g g$$

$$Q \quad (-1) \quad C_g \cdot C_g = (C_g)^2 = +1$$

C -parity violation in strong interaction
 $\Rightarrow$ allowed.

$$J/\psi \rightarrow g g g \quad \text{Channel.}$$



$$J/\psi \rightarrow (\bar{q}q) + (Q\bar{Q})$$

$$Q, q = u, d, s$$

$$J/\psi \rightarrow g g g$$

$$Q \quad -1 \quad (-1)(-1)(-1)$$

lightest $Q\bar{Q}, \bar{Q}q$: π^+, π^-

$$J/\psi \rightarrow \pi^+ \pi^-$$

$$J \quad 1 \quad 0 \quad 0$$

$$I \quad 0 \quad 1 \quad 1$$

$$S \quad 1 \quad 0 \quad 0$$

$$G \quad (-1) \quad (-1) \quad (-1)$$

$$G = C \times R_2 = C e^{i\pi T_2}$$

$$G = (-1)^{I+S}$$

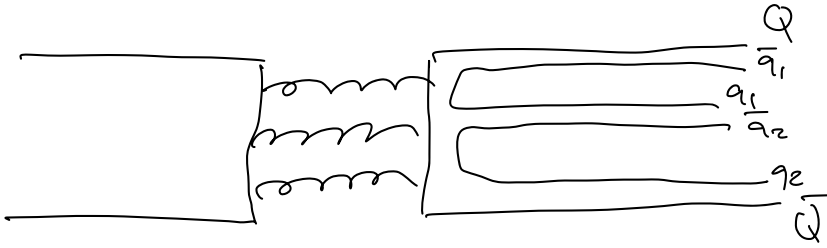
$\longleftrightarrow$
 G parity violation

$\Rightarrow J/\psi \rightarrow \pi^+ \pi^-$ not allowed.

$$\rho = \frac{u\bar{u} - d\bar{d}}{\sqrt{2}}$$

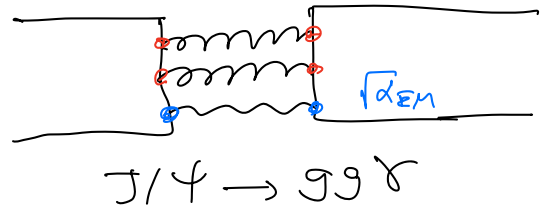
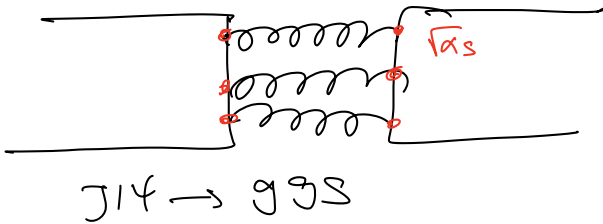
$$S=1$$

$$\Rightarrow J/\psi \rightarrow \pi \rho \rightarrow \pi^+ \pi^- \pi^0$$



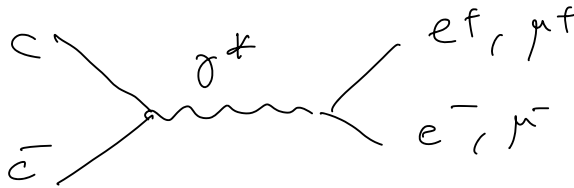
J/ψ(1S) DECAY MODES

Mode	Fraction (Γ_i/Γ)	Scale factor/ Confidence level
Γ_1 hadrons	(87.7 $\pm$ 0.5) %	
Γ_2 virtual $\gamma \rightarrow$ hadrons	(13.50 $\pm$ 0.30) %	
Γ_3 ggg	(64.1 $\pm$ 1.0) %	
Γ_4 γgg	(8.8 $\pm$ 1.1) %	
Γ_5 e^+e^-	(5.971 $\pm$ 0.032) %	
Γ_6 $e^+e^-\gamma$	[a] (8.8 $\pm$ 1.4) $\times 10^{-3}$	
Γ_7 $\mu^+\mu^-$	(5.961 $\pm$ 0.033) %	



$$\frac{\Gamma(J/\psi \rightarrow g g \gamma)}{\Gamma(J/\psi \rightarrow g g g)} = \frac{\sqrt{\alpha_{EM}} \cdot \sqrt{\alpha_{EM}}}{\sqrt{\alpha_s} \sqrt{\alpha_s}}$$

$$J/\psi \rightarrow e^+ e^-$$



$$\psi(2S) : c\bar{c} \text{ resonance} \quad m = 3.7 \text{ GeV} \quad \Gamma = 294 \text{ keV}$$

ψ(2S) DECAY MODES

Mode	Fraction (Γ_i/Γ)	Scale factor/ Confidence level
Γ_1 hadrons	(97.85 $\pm$ 0.13) %	
Γ_2 virtual $\gamma \rightarrow$ hadrons	(1.73 $\pm$ 0.14) %	S=1.5
Γ_3 ggg	(10.6 $\pm$ 1.6) %	
Γ_4 γgg	(1.03 $\pm$ 0.29) %	
Γ_5 light hadrons	(15.4 $\pm$ 1.5) %	
Γ_6 K_S^0 anything	(16.0 $\pm$ 1.1) %	
Γ_7 e^+e^-	(7.93 $\pm$ 0.17) $\times 10^{-3}$	
Γ_8 $\mu^+\mu^-$	(8.0 $\pm$ 0.6) $\times 10^{-3}$	
Γ_9 $\tau^+\tau^-$	(3.1 $\pm$ 0.4) $\times 10^{-3}$	

Decays into J/ψ(1S) and anything

Γ_{10} J/ψ(1S) anything	(61.4 $\pm$ 0.6) %
Γ_{11} J/ψ(1S) neutrals	(25.38 $\pm$ 0.32) %
Γ_{12} J/ψ(1S) $\pi^+\pi^-$	(34.68 $\pm$ 0.30) %
Γ_{13} J/ψ(1S) $\pi^0\pi^0$	(18.24 $\pm$ 0.31) %
Γ_{14} J/ψ(1S) η	(3.37 $\pm$ 0.05) %
Γ_{15} J/ψ(1S) π^0	(1.268 $\pm$ 0.032) $\times 10^{-3}$

unbalanced events: $e^+e^- \rightarrow e^+\mu^- + \text{missing stuff}$

$$e^+e^- \rightarrow \tau^+ \tau^-$$

$$\tau^- \rightarrow \mu^- \bar{\nu}_\mu \nu_\tau$$

$$\tau^+ \rightarrow e^+ \nu_e \bar{\nu}_\tau$$

$$\begin{pmatrix} \tau^- \\ \nu_e \end{pmatrix} L_\tau = -1$$

$$\begin{pmatrix} \bar{\nu}_\tau \\ \tau^+ \end{pmatrix} L_\tau = +1$$

$$e^+\mu^-$$

$$e^-\mu^+$$

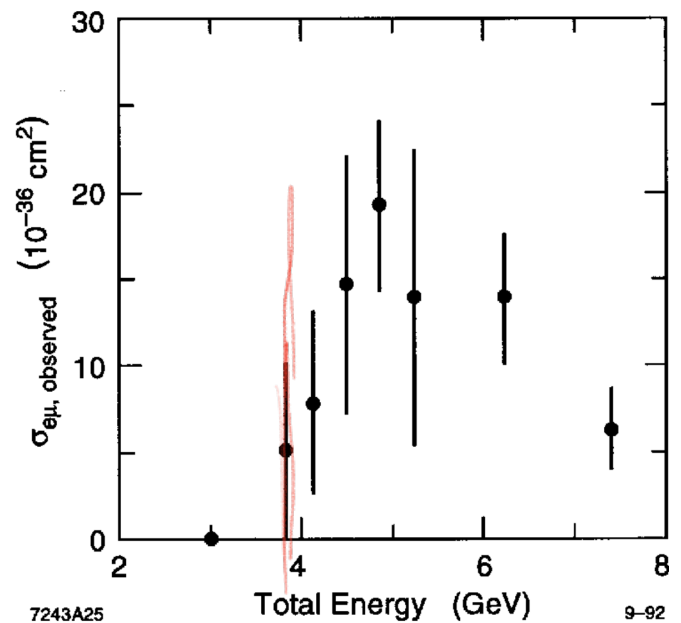
$$e^+e^-$$

$$\mu^+\mu^-$$

$$\#(e^+e^- \rightarrow e^+\mu^-)$$

as function of $\sqrt{s}$

threshold at $2m_\tau$



b Quark

Fermilab @ 1977

$$e^+e^- \rightarrow \gamma^* \rightarrow b\bar{b} \quad \text{requires knowledge of } \sqrt{s} \approx 2m_b$$

hadron production;

$$p + Cu(pT) \rightarrow \underbrace{(\mu^+\mu^-)}_{\text{resonance}} + X$$

$$p + p \rightarrow X b \bar{b} + Y + Z$$

$$\hookrightarrow \mu^+\mu^-$$

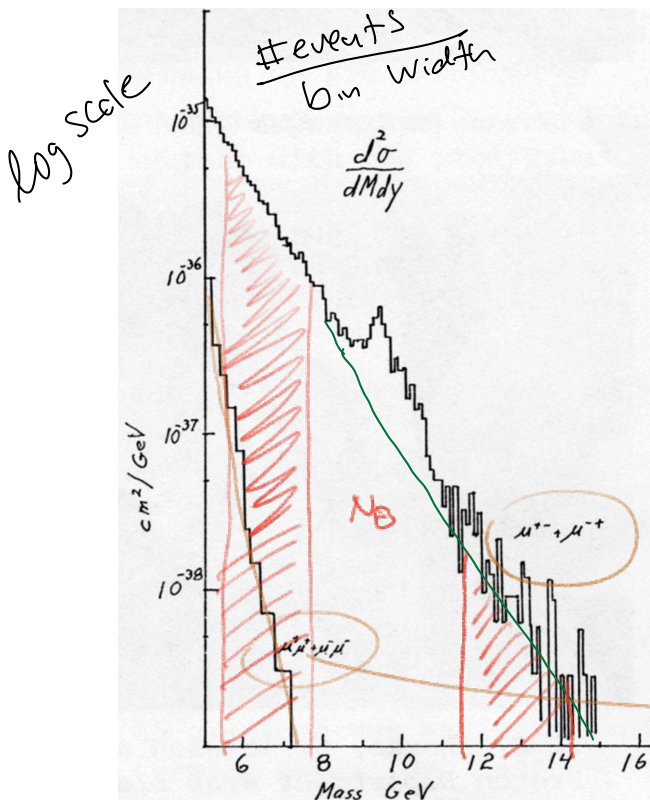
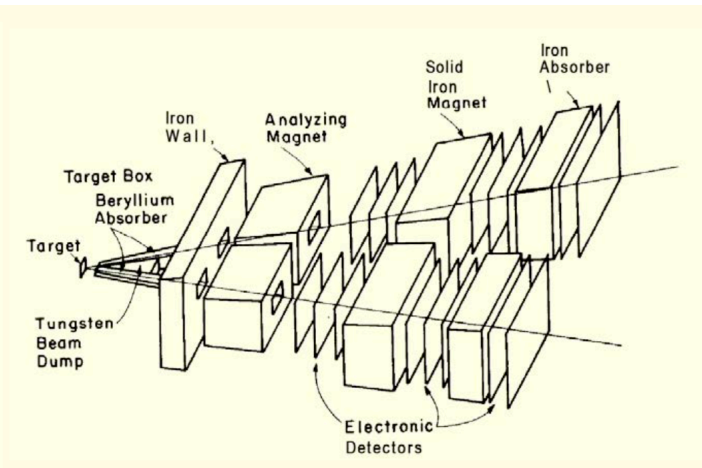
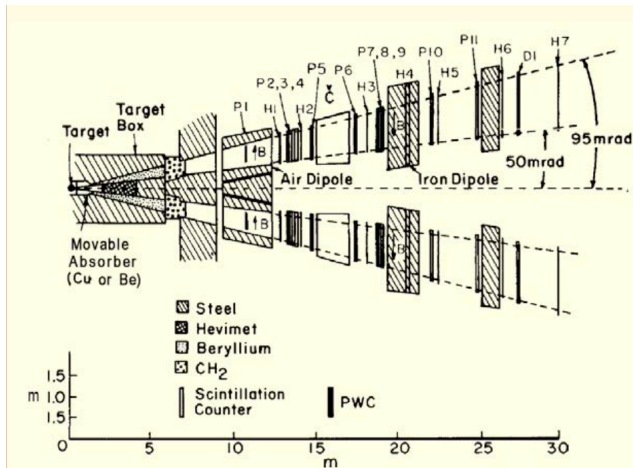
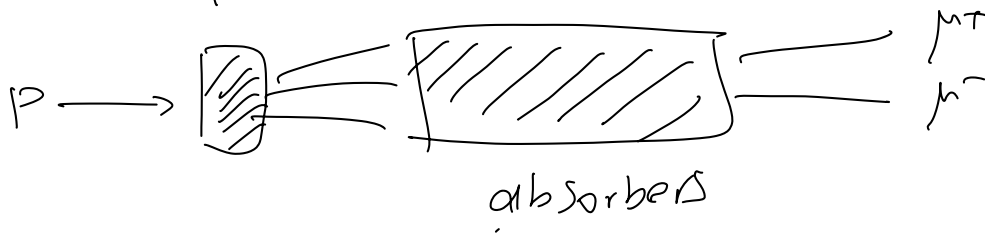
look for peak in invariant mass $\mu^+\mu^-$

$$p + Cu(pT) \rightarrow \pi^\pm K^\mp K^\pm + K^0 + p + \bar{p} + \dots$$

$\sqrt{s} \approx 10 \text{ GeV}$ π, μ have similar mass.

$$E = p^2 + m^2 \quad m_\pi \approx m_\mu$$

$\Rightarrow$ Stop all pions. $\Rightarrow$ only $\mu^\pm$ can survive



measure $\vec{p}_{\mu^+}, \vec{p}_{\mu^-}$ (from spectrometer)

$$E^2 = m_\mu^2 + p^2$$

$\Rightarrow$ compute $m_{\mu^+\mu^-} = \sqrt{(p_{\mu^+} + p_{\mu^-})^2}$

$$= (E_{\mu^+} + E_{\mu^-})^2 - (\vec{p}_{\mu^+} + \vec{p}_{\mu^-})^2$$

$p + \text{target} \rightarrow \mu^+ \mu^+ + X$
 $\mu^- \mu^- + X$
 Same-sign muons

If no real particle $\gamma^* \rightarrow \mu^+ \mu^+ \Rightarrow$ expect pure Background.

$\Rightarrow$ estimate shape of background distribution.

Background $A e^{-\alpha m_{\mu\mu}} \Rightarrow$ gives α, A .

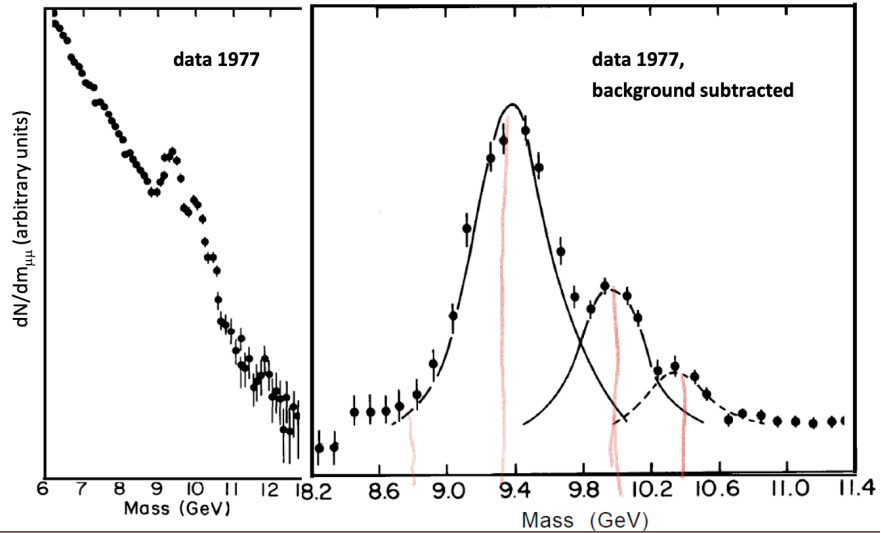
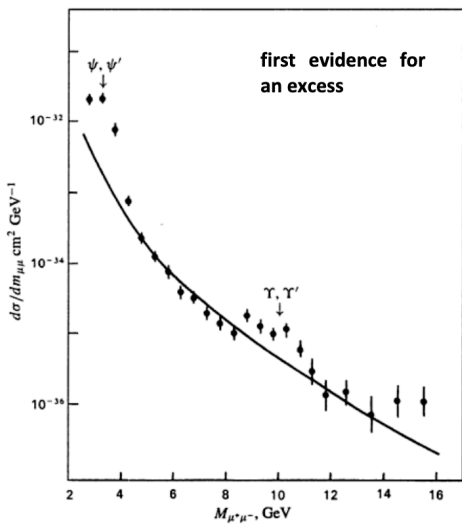
Fit data in $m_{\mu\mu} \in [11, 14] + [5.5, 7.5]$

$$N_B^{est} = \int_8^{10} A e^{-\alpha m_{\mu\mu}} dm$$

estimate of background
in signal region.

$$\hat{N}_S = N_{tot} - N_B^{est}$$

↪ # events in [8, 10]



$$A e^{-\alpha m} + S_1 e^{-\frac{(m-m_1)^2}{2\sigma_1^2}} + S_2 e^{-\frac{(m-m_2)^2}{2\sigma_2^2}} + S_3 e^{-\frac{(m-m_3)^2}{2\sigma_3^2}}$$

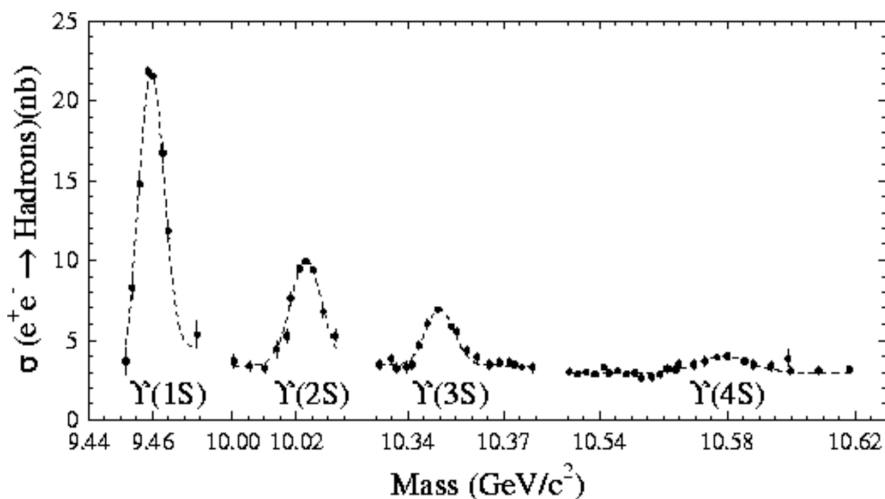
Bottomonium ($b\bar{b}$) resonance

Upsilon ($1S, 2S, 3S$)

⇒ e^+e^- machine $\sqrt{s}$ chosen to study each resonance

e^+e^- to study $b\bar{b}$ resonance.

$$\sqrt{s} \geq 9.4 \text{ GeV} \quad e^+e^- \rightarrow e^+e^-, \mu^+\mu^-, \text{hadrons}, \tau^+\tau^-$$



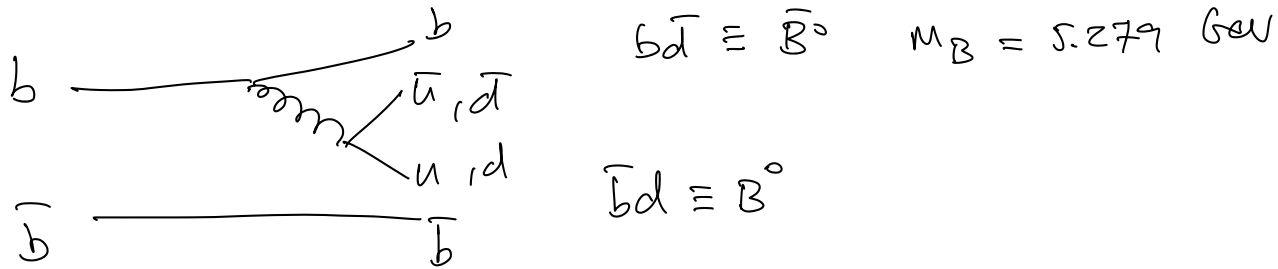
e^+e^- at Cornell University.
CESR (CLEO experiment)

PEP-II machine @ SLAC
BaBar exp.

KEK machine in Japan.
Belle (II) exp

$$\Upsilon(1S) \quad \Gamma: 52 \text{ keV}$$

$$\Upsilon(4S) \quad \Gamma = 20.5 \text{ MeV}$$



$$m(Y(1S)) < 2m_B$$

$Y(1S) \rightarrow ggs$ allowed.

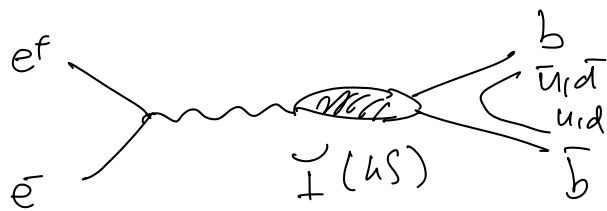
$$M = 10.58 \text{ GeV} > 2m_B$$

$Y(1S)$ DECAY MODES

Mode	Fraction (Γ_i/Γ)	Scale factor/ Confidence level
$\Gamma_1 \quad \tau^+ \tau^-$	(2.60 $\pm$ 0.10) %	
$\Gamma_2 \quad e^+ e^-$	(2.39 $\pm$ 0.08) %	
$\Gamma_3 \quad \mu^+ \mu^-$	(2.48 $\pm$ 0.04) %	
Hadronic decays		
$\Gamma_4 \quad ggg$	(81.7 $\pm$ 0.7) %	
$\Gamma_5 \quad \gamma gg$	(2.2 $\pm$ 0.6) %	
$\Gamma_6 \quad \eta'(958) \text{ anything}$	(2.94 $\pm$ 0.24) %	
$\Gamma_7 \quad J/\psi(1S) \text{ anything}$	(5.4 $\pm$ 0.4) $\times 10^{-4}$	S=1.4
$\Gamma_8 \quad J/\psi(1S) \eta_c$	< 2.2 $\times 10^{-6}$	CL=90%
$\Gamma_9 \quad J/\psi(1S) \chi_{c0}$	< 3.4 $\times 10^{-6}$	CL=90%
$\Gamma_{10} \quad J/\psi(1S) \chi_{c1}$	(3.9 $\pm$ 1.2) $\times 10^{-6}$	
$\Gamma_{11} \quad J/\psi(1S) \chi_{c2}$	< 1.4 $\times 10^{-6}$	CL=90%

$Y(4S)$ DECAY MODES

Mode	Fraction (Γ_i/Γ)	Confidence level
$\Gamma_1 \quad B \bar{B}$	> 96 %	95%
$\Gamma_2 \quad B^+ B^-$	(51.4 $\pm$ 0.6) %	
$\Gamma_3 \quad D_s^+ \text{ anything} + \text{c.c.}$	(17.8 $\pm$ 2.6) %	
$\Gamma_4 \quad B^0 \bar{B}^0$	(48.6 $\pm$ 0.6) %	
$\Gamma_5 \quad J/\psi K_S^0 + (J/\psi, \eta_c) K_S^0$	< 4 $\times 10^{-7}$	90%
$\Gamma_6 \quad \text{non-} B \bar{B}$	< 4 %	95%
$\Gamma_7 \quad e^+ e^-$	(1.57 $\pm$ 0.08) $\times 10^{-5}$	
$\Gamma_8 \quad \rho^+ \rho^-$	< 5.7 $\times 10^{-6}$	90%
$\Gamma_9 \quad K^*(892)^0 \bar{K}^0$	< 2.0 $\times 10^{-6}$	90%
$\Gamma_{10} \quad J/\psi(1S) \text{ anything}$	< 1.9 $\times 10^{-4}$	95%
$\Gamma_{11} \quad D^{*+} \text{ anything} + \text{c.c.}$	< 7.4 %	90%
$\Gamma_{12} \quad \phi \text{ anything}$	(7.1 $\pm$ 0.6) %	
$\Gamma_{13} \quad \phi \eta$	< 1.8 $\times 10^{-6}$	90%
$\Gamma_{14} \quad \phi \eta'$	< 4.3 $\times 10^{-6}$	90%
$\Gamma_{15} \quad \rho \eta$	< 1.3 $\times 10^{-6}$	90%



$$Y(4S) \rightarrow B^0 \bar{B}^0, B^+ B^-$$

$$e^+ e^- \text{ with } \sqrt{s} \approx m_{Y(4S)} = 10.58 \text{ GeV} \quad B\text{-Factory}$$

$e^+ e^- \rightarrow B^0 \bar{B}^0, B^+ B^-$ } dominantly because a resonance.
also $u\bar{u}, d\bar{d}, s\bar{s}$

$$B \xleftrightarrow{Y(4S)} \bar{B}$$

$$Q = 10.58 - 2 \times 5.279 \text{ very small.}$$

$$P_B^* \approx 200 \text{ MeV}$$

$$\beta \approx \frac{P}{m} = \frac{200}{5279} \approx 4\%$$

$$\frac{P}{E} = \frac{200}{\sqrt{200^2 + 5279^2}} \approx 0.04.$$

$\Rightarrow B, \bar{B}$ almost at rest

$$B_s = \bar{b}$$

